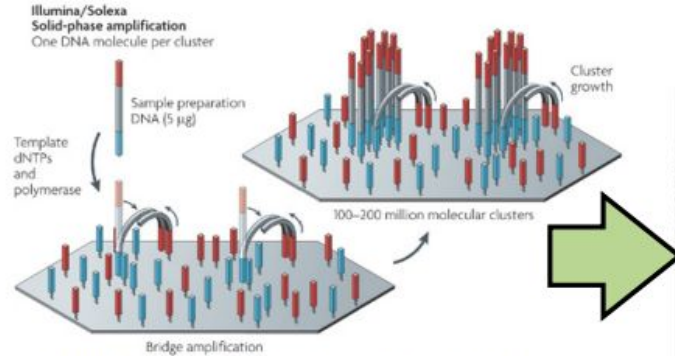


Pre-processing and normalization

Jeff Leek

@jtleek

www.jtleek.com



Source: Metzker ML. Sequencing technologies - the next generation. Nat Rev Genet. 2010

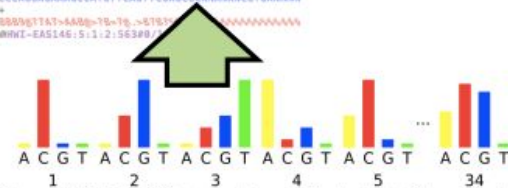
```

@HWI-EAS146:5:1:1:961#0/1
TCCGAGGCCAACCGAGCTCCCGCGCTGNNNNNNNNNNNNNN
+
BBB@-A7B@;@BBBBA-BA-AAAAAAAAAAAAAAAAAAAA
@HWI-EAS146:5:1:1:1595#0/1
TCAGGAAGCAGGAGAGCTGGTCAGCAGNNNNNNNNNNNNNN
+
BBB@-;BAA-@BB@-1>NNNNNNNNNNNNNNNNNNNNNN
@HWI-EAS146:5:1:1:1848#0/1
CTGGACTGCATCCTACCACCACTCTCCAAANNNNNNNNNN
+
A-B7B7>BB-A>79<:;>747NNNNNNNNNNNNNNNNNN
@HWI-EAS146:5:1:1:1719#0/1
CTCCTCTCAAGGTCCCGAAGCAGCAGCAANNNNANTNCTNNN
+
BBCCCCCDBCB7CBC=7>=<=>BCBCBNNNNNNNNNNNN
@HWI-EAS146:5:1:1:1719#0/1
CAGGATCTGGTTTATTGTATCACTCCGCTCNNGNNTNAGNNNN
+
BCC7<=<B=7BBS=ABA7BGBBBA7B7BNNNNNNNNNNNN
@HWI-EAS146:5:1:2:947#0/1
CCAGGAGAGGCCATGTTCACTTCGAGCGCNHANCCTGANNNN
+
BBB@TTAT>AAB@>7B=7B>=>STB>
@HWI-EAS146:5:1:2:563#0/1

```

name
sequence
quality scores

x 100s of
millions



Source: Whiteford et al. Swift: primary data analysis for the Illumina Solexa sequencing platform. Bioinformatics. 2009

Illumina/Solexa — Reversible terminators

Incorporate all four nucleotides, each label with a different dye

Wash, four-colour imaging

Cleave dye and terminating groups, wash

Repeat cycles

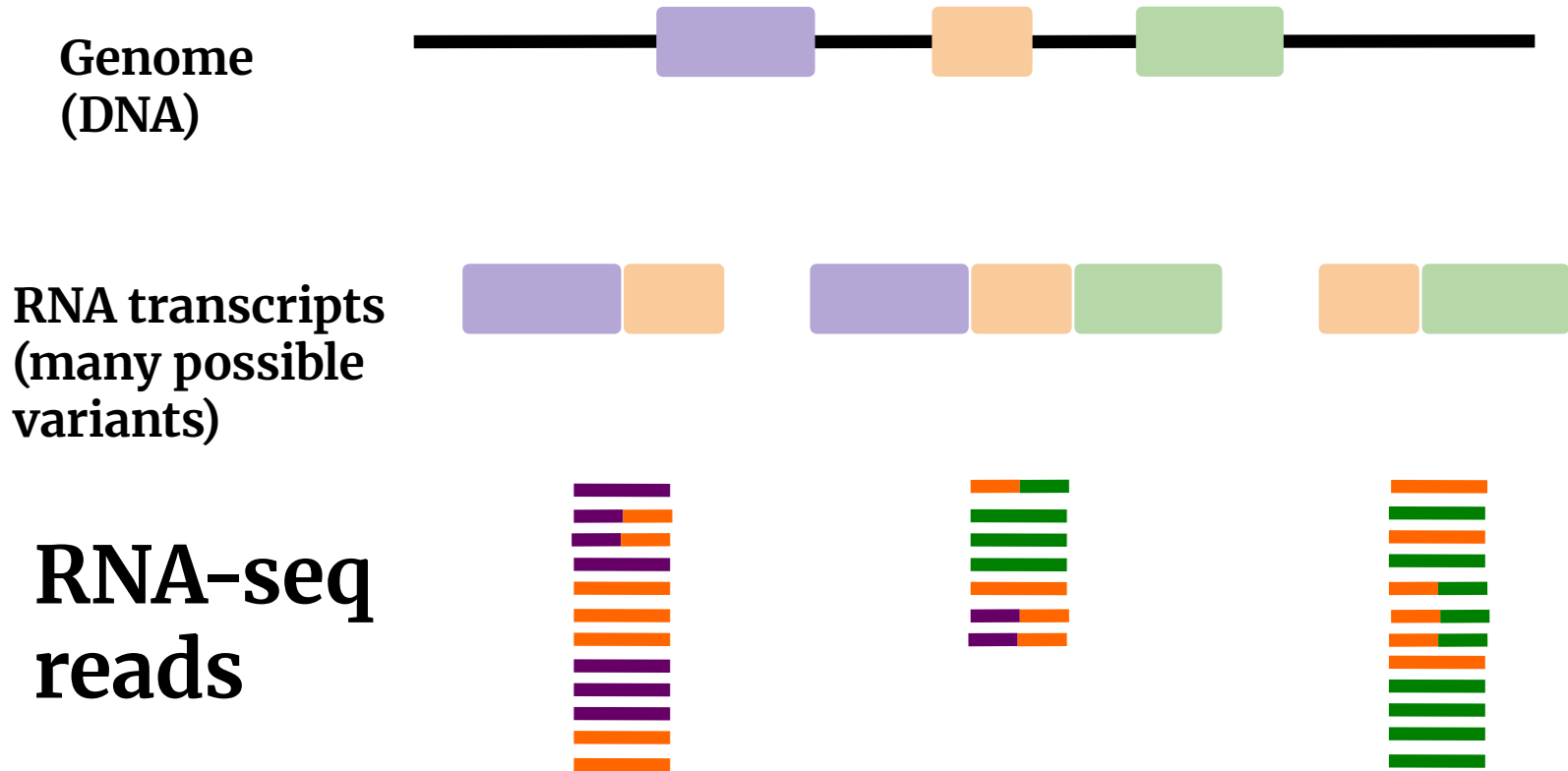


Source: Metzker ML. Sequencing technologies - the next generation. Nat Rev Genet. 2010

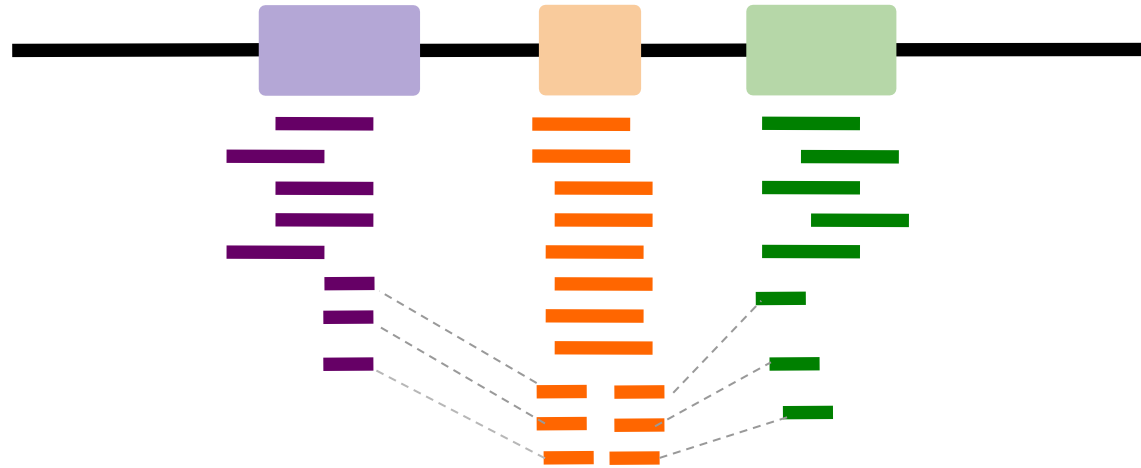
Preprocessing

Convert raw data to “processed”

Try to remove technological artifacts

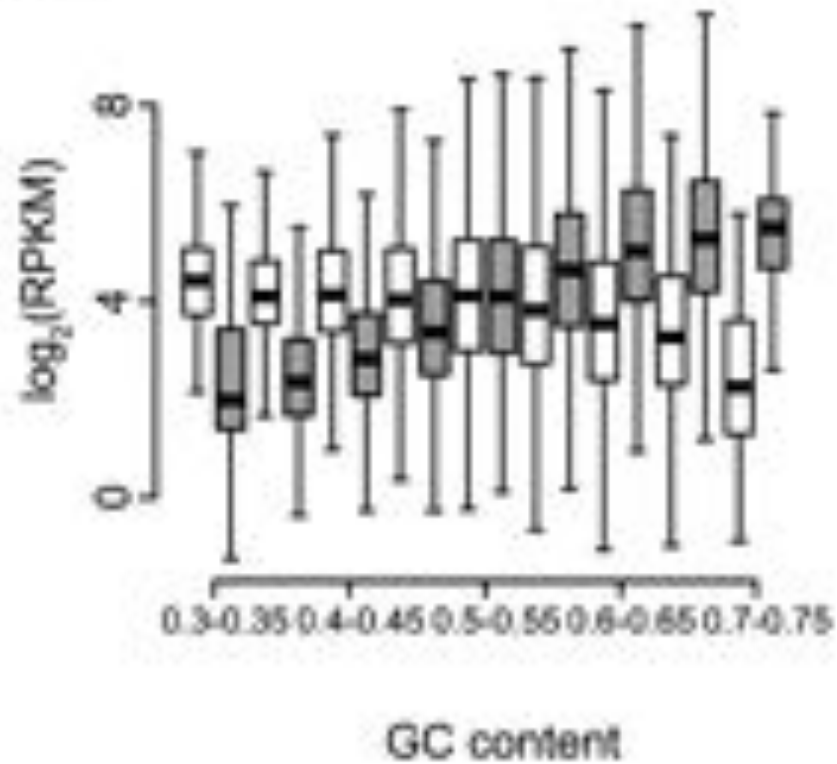


Genome
(DNA)

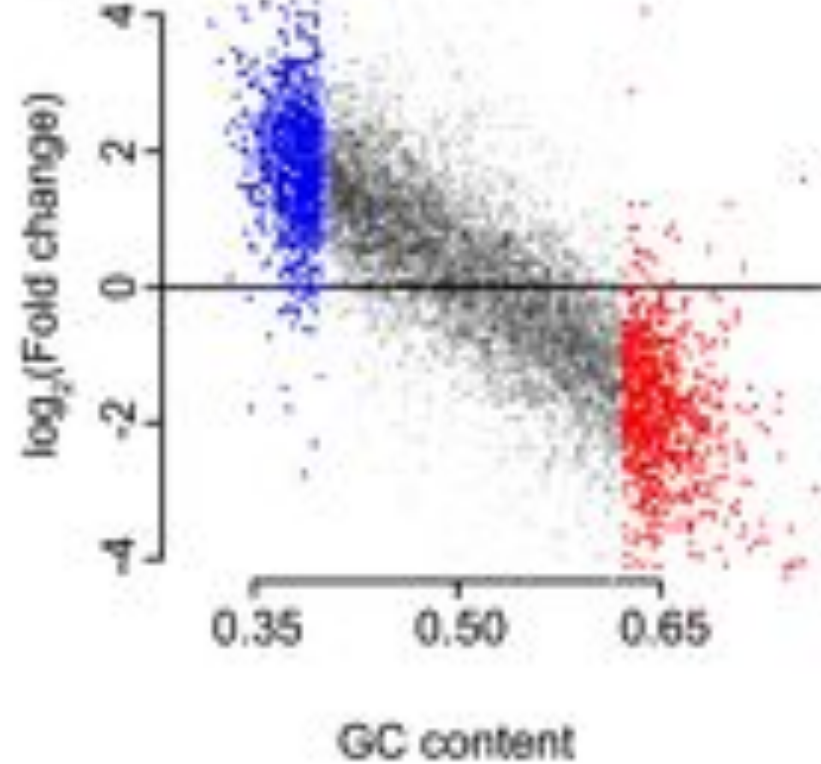


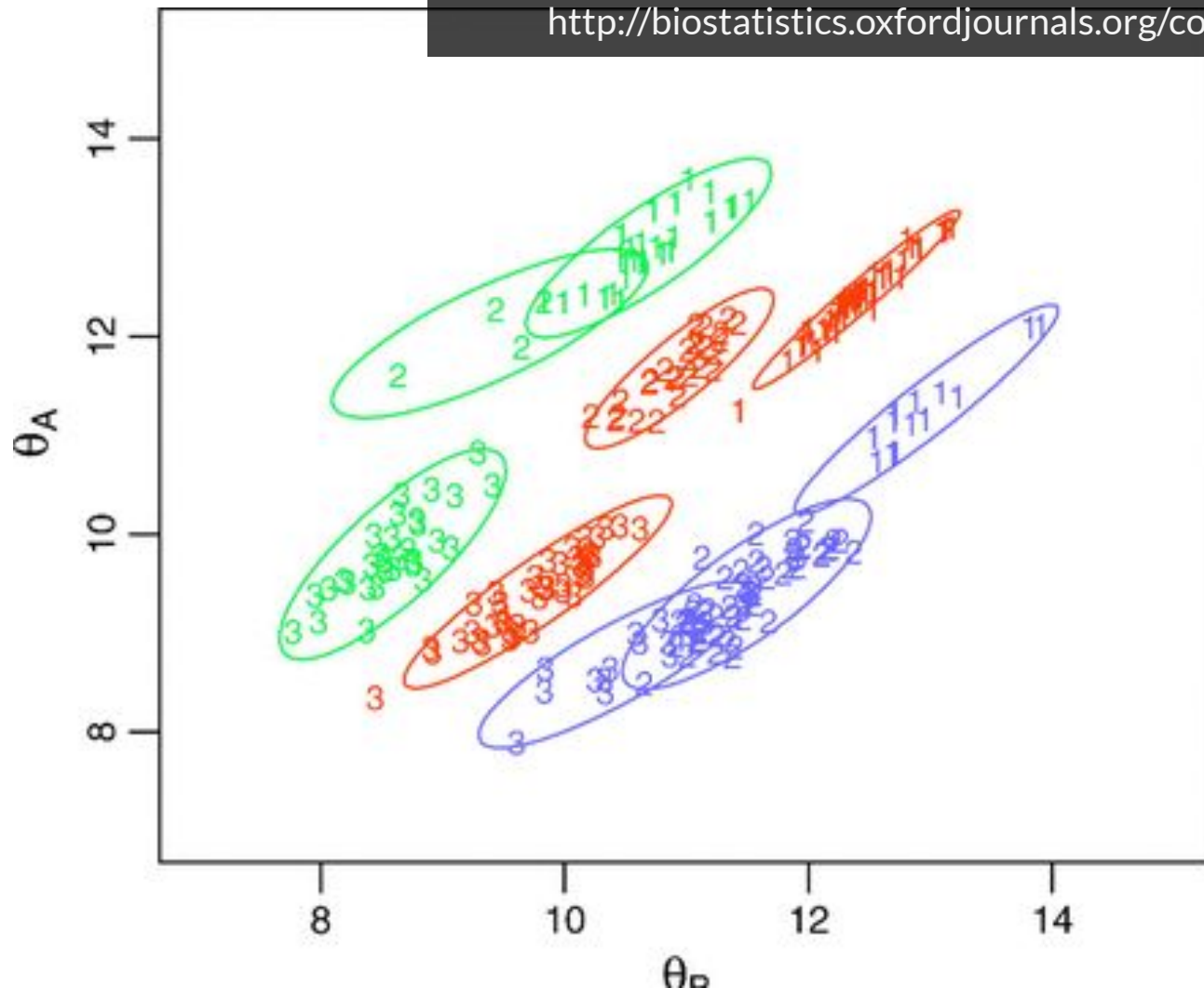
expression = 24

(b)



(c)





Normalization

Remove technological biases

Make samples comparable

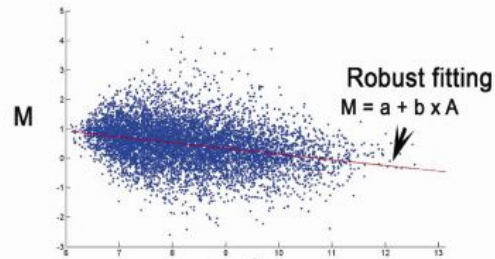
Sample 1

Peak Coordinates
Read Coordinates

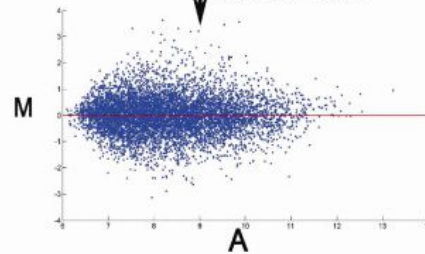
Sample 2

Peak Coordinates
Read Coordinates

MA plot of common peaks



A
Normalization



Quantile normalization

Most common technique

Bulk distributions exactly the same

Raw data

2	4	4	5
5	14	4	7
4	8	6	9
3	8	5	8
3	9	3	5

**Order values
within each sample
(or column)**

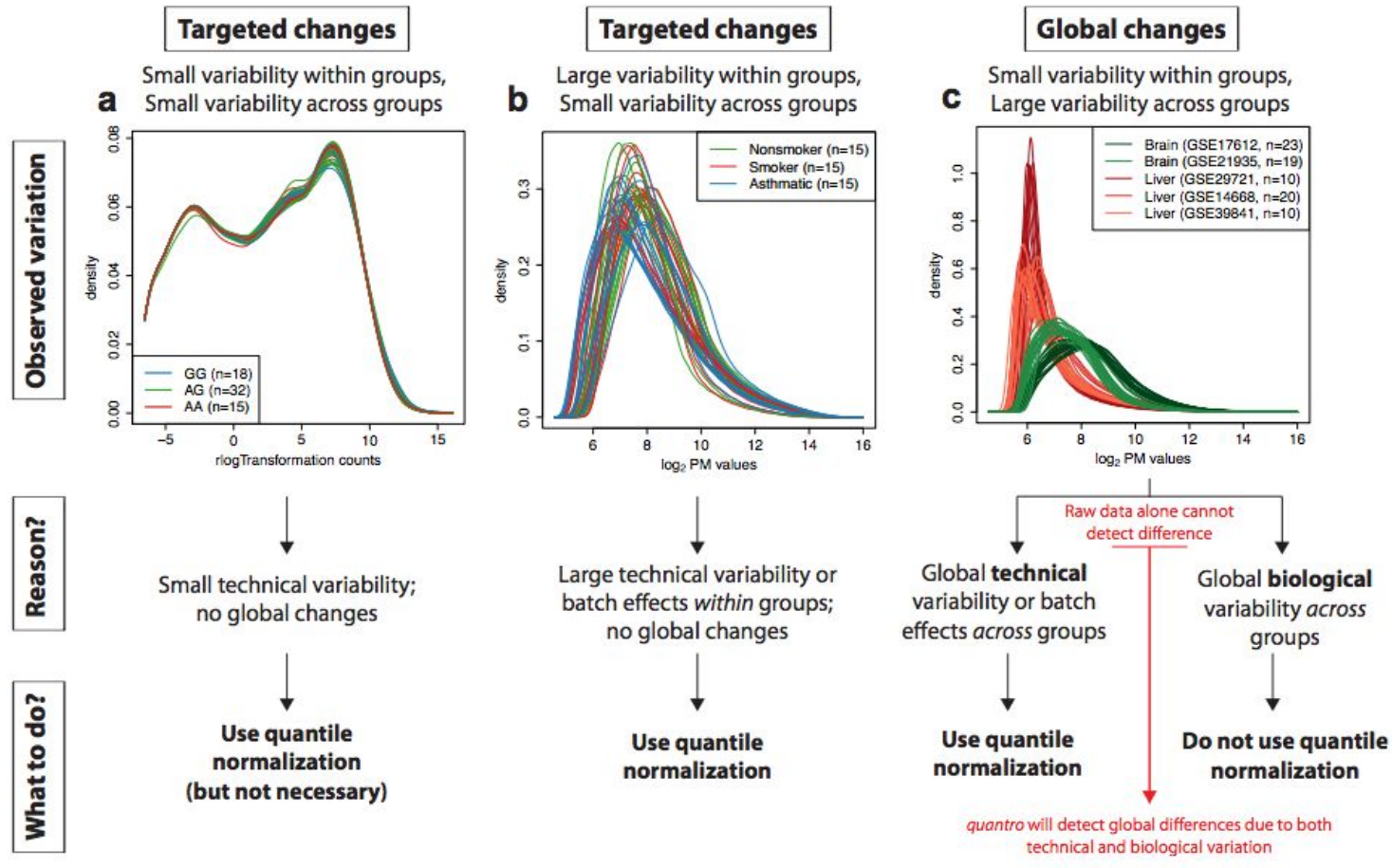
2	4	3	5
3	8	4	5
3	8	4	7
4	9	5	8
5	14	6	9

**Average across rows
and substitute value
with average**

3.5	3.5	3.5	3.5
5.0	5.0	5.0	5.0
5.5	5.5	5.5	5.5
6.5	6.5	6.5	6.5
8.5	8.5	8.5	8.5

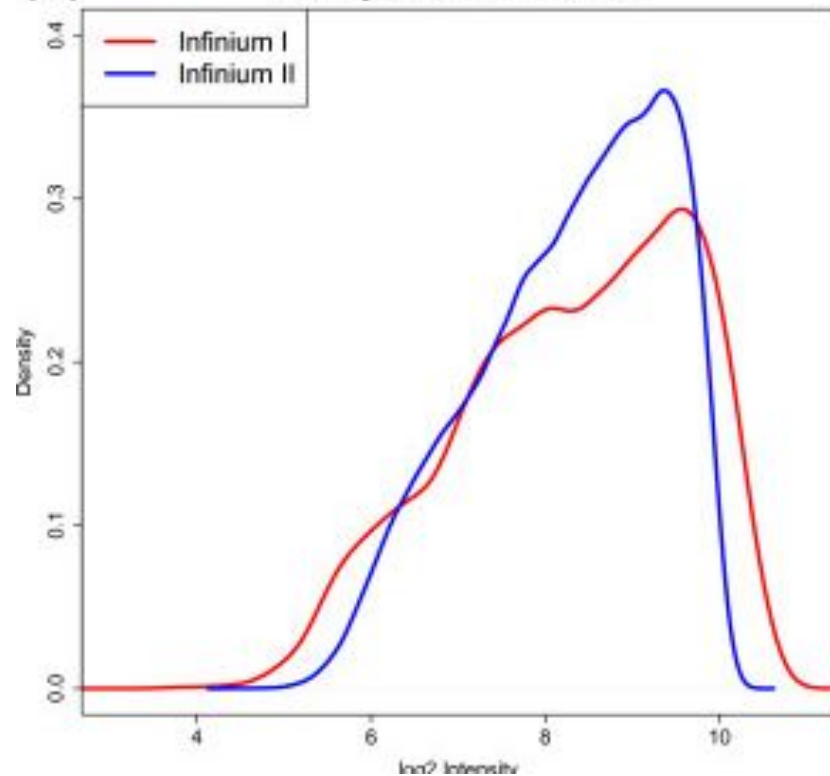
**Re-order averaged
values in original
order**

3.5	3.5	5.0	5.0
8.5	8.5	5.5	5.5
6.5	5.0	8.5	8.5
5.0	5.5	6.5	6.5
5.5	6.5	3.5	3.5

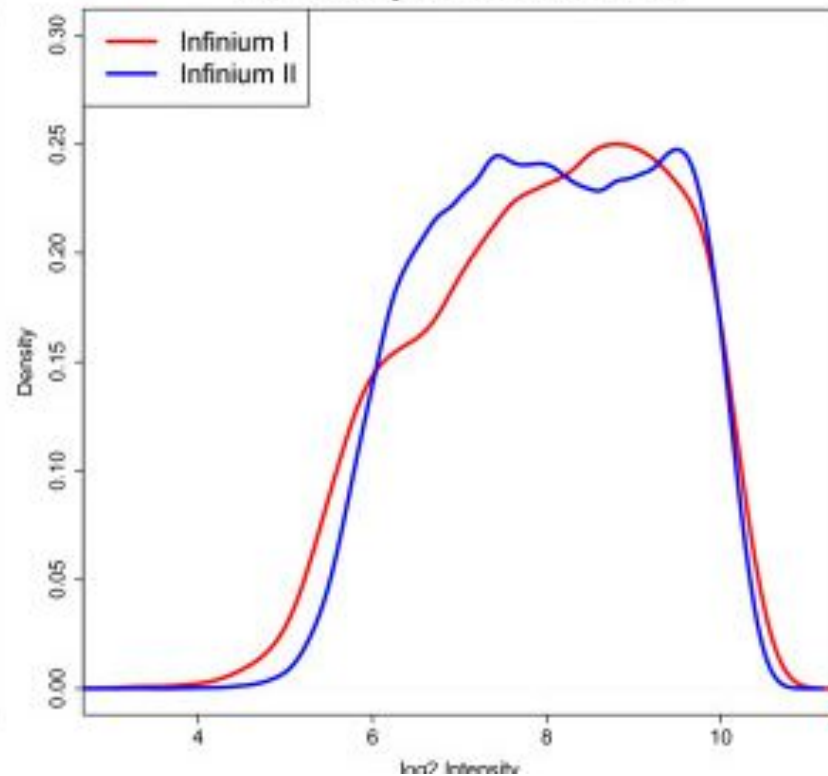


(a)

Methylated Channel



Unmethylated Channel



Notes and further reading

- Preprocessing and normalization are highly platform/problem dependent
- In general check to make sure there aren't bulk differences between samples, especially due to technology
- Bioconductor workflows are a good place to start:

<https://www.bioconductor.org/help/workflows/>

“ First, researchers starting out in genomics must keep in mind that interesting outliers — that is, results that deviate significantly from the sample — will inevitably contain a plethora of experimental or analytical artefacts. ”